

Unit - III

Matrix Decomposition

Q-R Factorization:-

If A is an $m \times n$ matrix with linearly independent columns then it can be factored as

$$A = QR$$

Where Q is an $m \times n$ orthogonal matrix (where $Q^T = Q^{-1}$) whose columns form an orthonormal basis for $\text{Col}(A)$.
 R is an $n \times n$ upper triangular matrix with positive entries on the diagonals.

Working principle:

1. Let the columns of $A = \{u_1, u_2, \dots, u_n\}$ which forms the basis of $\text{colsp}(A)$.
2. Construct an orthonormal basis $\{\hat{v}_1, \hat{v}_2, \dots, \hat{v}_n\}$ from $\{u_1, u_2, u_3, \dots, u_n\}$ by Gram-Schmidt.
3. Construct $Q = [\hat{v}_1, \hat{v}_2, \dots, \hat{v}_n]$.
4. $A = QR \Rightarrow Q^T A = R$ ($\because Q$ is orthogonal, $Q^{-1} = Q^T$).
5. The required factorization is $A = QR$.

Find the QR decomposition of matrix $\begin{pmatrix} 1 & -1 \\ -1 & 0 \end{pmatrix}$

$$\text{Let } u_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad u_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

Applying Gram-Schmidt method
 $v_1 = u_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$v_2 = u_2 - \frac{u_2 \cdot v_1}{v_1 \cdot v_1} v_1$$

$$= \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \frac{(-1)}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$v_2 = \begin{pmatrix} -\frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$$

Normalizing the vectors

$$\hat{v}_1 = \frac{1}{\|v_1\|} v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\hat{v}_2 = \frac{1}{\|v_2\|} v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -\frac{1}{2} \\ -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

Construct

The matrix Q is

$$Q = [\hat{v}_1 \hat{v}_2] = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

To construct R

$$A = QR$$

$$R = Q^{-1}A$$

$$\Rightarrow R = Q^T A \quad [\because \text{Since } Q \text{ is an orthogonal matrix}]$$

$$R = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$$

$\therefore$ The required QR decomposition is

$$R = \begin{bmatrix} \sqrt{2} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

2. Find QR for Matrix $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \\ 1 & -1 \end{pmatrix}$

$$\text{Let } A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \\ 1 & -1 \end{bmatrix}$$

$$\text{Let } u_1 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \quad u_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

~~Let QR~~

Applying Gram-Schmidt method

$$v_1 = u_1 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

$$v_2 = u_2 - \frac{u_2 \cdot v_1}{v_1 \cdot v_1} v_1$$

$$= \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} - \frac{(-1)}{2} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

$$v_2 = \begin{pmatrix} 1 \\ -\frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$$

Normalizing the vectors

$$Q_1 = \frac{1}{\|v_1\|} v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\hat{v}_2 = \frac{1}{\|v_2\|} v_2 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{3} \\ -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \end{pmatrix}$$

Constructing Q

$$Q = [\hat{v}_1 \hat{v}_2] = \begin{bmatrix} 0 & \frac{\sqrt{2}}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \end{bmatrix}$$

Finding R

$$A = QR$$

$$R = Q^T A$$

$$= \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{\sqrt{2}}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{bmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \\ 1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} + \frac{-1}{\sqrt{6}} & \frac{\sqrt{2}}{\sqrt{3}} + \frac{1}{\sqrt{6}} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{2}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{0}{\sqrt{6}} & \frac{3}{\sqrt{6}} \end{pmatrix}$$

$$R = \begin{pmatrix} \sqrt{2} & 0 \\ 0 & \frac{\sqrt{3}}{\sqrt{2}} \end{pmatrix}$$

3 QR for $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

Let $u_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ $u_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ $u_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

Gram-Schmidt method

$$v_1 = u_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$V_2 = u_2 - \frac{u_2 \cdot V_1}{V_1 \cdot V_1} V_1$$

$$= 4 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$V_2 = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{pmatrix}$$

$$V_3 = u_3 - \frac{V_2 \cdot u_3}{V_2 \cdot V_2} V_2 - \frac{V_1 \cdot u_3}{V_1 \cdot V_1} V_1$$

$$V_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{\frac{1}{2}}{\frac{3}{2}} \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$V_3 = \begin{pmatrix} -\frac{2}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{pmatrix}$$

Normalizing the vector

$$\hat{V}_1 = \frac{1}{\|V_1\|} V_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$$

$$\hat{V}_2 = \frac{1}{\|V_2\|} V_2 = \frac{1}{\sqrt{\frac{3}{2}}} \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{\sqrt{2}}{\sqrt{3}} \end{pmatrix}$$

$$\hat{V}_3 = \frac{1}{\|V_3\|} V_3 = \frac{1}{\frac{2}{\sqrt{3}}} \begin{pmatrix} -\frac{2}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{pmatrix} = \begin{pmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{\sqrt{2}}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

$$-\frac{1}{\sqrt{6}} + \frac{2\sqrt{2}}{\sqrt{6}} = \frac{2}{\sqrt{6}} = \frac{\sqrt{2}}{\sqrt{3}}$$

Find $R =$

$$R = Q^T A$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$R = \begin{bmatrix} \frac{2}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{\sqrt{2}}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}$$

$$4 \quad A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$u_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad u_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad u_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

QR factorization

Using Gram-Schmidt

$$v_1 = u_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$v_2 = u_2 - \frac{u_2 \cdot v_1}{v_1 \cdot v_1} v_1$$

$$= \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix} - \frac{3}{4} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$v_2 = \begin{pmatrix} -\frac{3}{4} \\ \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{4} \end{pmatrix}$$

$$v_3 = u_3 - \frac{u_3 \cdot v_2}{v_2 \cdot v_2} v_2 - \frac{u_3 \cdot v_1}{v_1 \cdot v_1} v_1$$

$$= \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} - \frac{\frac{1}{2}}{\frac{3}{4}} \begin{pmatrix} -3 \\ 4 \\ 1 \\ 4 \end{pmatrix} - \frac{2}{4} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} - \frac{2}{3} \begin{pmatrix} -3 \\ 4 \\ 1 \\ 4 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$v_3 = \begin{pmatrix} 0 \\ -\frac{2}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix}$$

Normalization of the vectors

$$\hat{v}_1 = \frac{1}{\|v_1\|} v_1 = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

$$\hat{v}_2 = \frac{1}{\|v_2\|} v_2 = \frac{1}{\frac{\sqrt{3}}{2}} \begin{pmatrix} -3 \\ 4 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} \\ \frac{1}{2\sqrt{3}} \\ \frac{1}{2\sqrt{3}} \\ \frac{1}{2\sqrt{3}} \end{pmatrix}$$

$$\hat{v}_3 = \frac{1}{\|v_3\|} v_3 = \frac{1}{\frac{2}{\sqrt{3}}} \begin{pmatrix} 0 \\ -\frac{2}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{\sqrt{3}}{2} \\ \frac{1}{2\sqrt{3}} \\ \frac{1}{2\sqrt{3}} \end{pmatrix}$$

$$Q = \begin{bmatrix} \frac{1}{2} & \frac{-\sqrt{3}}{2} & -\cancel{0} \\ \frac{1}{2} & \frac{1}{2\sqrt{3}} & \frac{\sqrt{2}}{\sqrt{3}} \\ \frac{1}{2} & \frac{1}{2\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{2} & \frac{1}{2\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

Factorizing for R

$$R = Q^T A$$

$$R = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{-\sqrt{3}}{2} & \frac{1}{2\sqrt{3}} & \frac{1}{2\sqrt{3}} & \frac{1}{2\sqrt{3}} \\ 0 & \frac{\sqrt{2}}{3} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{\sqrt{2}}{3} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$R = \begin{bmatrix} 2 & \frac{3}{2} & 1 \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{\sqrt{3}} \\ 0 & 0 & \sqrt{\frac{2}{3}} \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}$$

Least square solution of an inconsistent system.

Consider a system $AX = b$ is inconsistent

When no such solution exists, it is required to find X such that AX is closest to b

The general least squares problems to find X such that $\|AX - b\|$ is minimum

Method of solution

Define error, $e = b - AX$

Purpose is to make the error as small as possible

to minimize: $\|b - AX\|^2$

Working rule

1. Given A and b , find A^T
2. Find $A^T A$ and $A^T b$
3. Substitute these in the normal equation $A^T A X = A^T b$
(This is called a normal equation)
4. Solve the above system by any known method

* Solve $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$ $b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$

$$A^T = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

$$\text{Let } X = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\text{Sub in } A^T A X = A^T b$$

$$\begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

$$17x + y = 19$$

$$x + 5y = 11$$

$$x = 1$$

$$y = 2$$

$$2. \quad A = \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$\text{Normal eqn } A^T A X = A^T b$$

$$A^T = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -3 & 3 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -3 & 3 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 6 & -11 \\ -11 & 22 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} -4 \\ 11 \end{bmatrix}$$

$$A^T A X = A^T b$$

$$\begin{bmatrix} 6 & -11 \\ -11 & 22 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -4 \\ 11 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{aligned} 6x - 11y &= -4 \\ -11x + 22y &= 11 \end{aligned}$$

$$x = 3 \quad y = 2$$

Therefore the least square solution is $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

$$\text{Ex. i) } A = \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 3 \end{bmatrix} \quad b = \begin{bmatrix} 5 \\ 8 \\ 1 \end{bmatrix}$$

$$\text{ii) } A = \begin{bmatrix} 1 & 3 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \quad b = \begin{bmatrix} -3 \\ -1 \\ 0 \\ 2 \\ 5 \\ 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 6 & 2 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 2 & 0 & 0 & 2 \end{bmatrix}$$

$$A^T B = \begin{bmatrix} 4 \\ -4 \\ 2 \\ 6 \end{bmatrix}$$

$$A^T A X = A^T B$$

$$\begin{bmatrix} 6 & 2 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 2 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \\ 2 \\ 6 \end{bmatrix}$$

$A^T A : A^T B$
The augmented matrix is

$$\left[\begin{array}{cccc|c} 6 & 2 & 2 & 2 & 4 \\ 2 & 2 & 0 & 0 & -4 \\ 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 0 & 2 & 6 \end{array} \right]$$

$$R_1 \leftrightarrow R_2$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$R_4 \rightarrow R_4 - R_1$$

$$\left[\begin{array}{cccc|c} 2 & 2 & 0 & 0 & -4 \\ 0 & -4 & 2 & 2 & 4 \\ 0 & -2 & 2 & 0 & 2 \\ 0 & -2 & 0 & 2 & 6 \end{array} \right]$$

$$R_3 \rightarrow 2R_3 - R_2$$

$$\left[\begin{array}{cccc|c} 2 & 2 & 0 & 0 & -4 \\ 0 & -4 & 2 & 2 & 4 \\ 0 & 0 & -1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 2 & 2 & 0 & 0 & -4 \\ 0 & 1 & -1 & 0 & -3 \\ 0 & 0 & -1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Rank(A) = 3 $\neq$ No. of variables = 4
 $\therefore$ consistent, infinite solutions

$\therefore t$ becomes free variable
 Let $t = k$

$\therefore$ 3 equations

$$2x_1 + 2x_2 = -4$$

$$x_2 - x_3 = -3$$

$$-x_3 + x_4 = 2$$

$$x_3 = x_4 - 2$$

$$x_2 = x_4 - 5$$

$$2x_1 + 2x_4 - 10 = -4$$

$$2x_1 =$$

$$x_1 + x_2 = -2$$

$$x_1 = -2 - x_4 + 5$$

$$x_1 = 3 - x_4$$

$\therefore$ The least square solution is

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 - x_4 \\ x_4 - 5 \\ x_4 - 2 \\ x_4 \end{bmatrix}$$

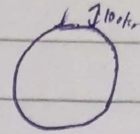
$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 3 \\ 8 \\ 2 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 8 \\ 2 \end{bmatrix} = \begin{bmatrix} 14 \\ 4 \\ 10 \end{bmatrix}$$

$$A^T A X = A^T b$$

$$\begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 14 \\ 4 \\ 10 \end{bmatrix}$$



Augmented matrix

$$\left[\begin{array}{ccc|c} 4 & 2 & 2 & 14 \\ 2 & 2 & 0 & 4 \\ 2 & 0 & 2 & 10 \end{array} \right]$$

$$R_1 \leftrightarrow R_2$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 2 & 2 & 0 & 4 \\ 0 & -2 & 2 & 6 \\ 0 & -2 & 2 & 6 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 2 & 2 & 0 & 4 \\ 0 & -2 & 2 & 6 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

z is free

Quadratic Form

A quadratic form is an expression in any number of variables, where each term is of degree 2. It is usually denoted by Q .

Eg:

$$Q = \underbrace{x_1^2}_{\text{square terms}} + \underbrace{2x_1x_2}_{\text{cross terms}} + \underbrace{4x_1x_3}_{\text{cross terms}} + \underbrace{3x_2^2}_{\text{square terms}} - \underbrace{3x_3^2}_{\text{square terms}}$$

— square terms

--- cross terms

Classifying Quadratic forms

$Q(x)$ is:

- positive definite if $Q(x) \geq 0$ for all $x \neq 0$
- negative definite if $Q(x) < 0$ for all $x \neq 0$
- Indefinite if $Q(x)$ assumes both positive and negative values.

The Quadratic form Q is defined on $\mathbb{R}^n$ whose value at a vector x in $\mathbb{R}^n$ can be computed by an expression:

$$Q(x) = x^T A x$$

where A is an $n \times n$ symmetric matrix called matrix of quadratic form.

$$A = \begin{bmatrix} \text{coeff}(x_1^2) & \frac{1}{2} \text{coeff}(x_1x_2) & \dots & \frac{1}{2} \text{coeff}(x_1x_n) \\ \frac{1}{2} \text{coeff}(x_1x_2) & \text{coeff}(x_2^2) & & \\ \vdots & & \ddots & \\ \frac{1}{2} \text{coeff}(x_1x_n) & & & \text{coeff}(x_n^2) \end{bmatrix}$$

Ex (above is Quadratic form)

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & 2 & -3 \end{bmatrix}$$

$Q(x)$ in R^3 let $Q(x) = 5x_1^2 + 3x_2^2 + 2x_3^2 + x_1x_2 + 8x_1x_3$

$$A = \begin{bmatrix} 5 & \frac{1}{2} & 4 \\ \frac{1}{2} & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Let $Q = X^T A X$ be a quadratic form. If A is diagonalizable, then there exists a matrix P such that $P^{-1} A P = D$, where D is a diagonal matrix.

Suppose P is orthogonal, then we have $P^{-1} = P^T$

$$A = P D P^{-1}$$

Then $Q = X^T A X$ can be written as

$$Q = X^T A X$$

$$\Rightarrow Q = Y^T D Y \text{ where } Y = P^T X$$

working rule:

1. write matrix A of quadratic form.
2. Find eigen values of A say $\lambda_1, \lambda_2, \dots, \lambda_n$
Express quadratic form in new variables i.e. Q
 $\lambda_1 y_1 + \lambda_2 y_2 + \dots + \lambda_n y_n$

$$\text{where } Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = P^T X$$

3. Mention the transformation Y columns of P are normalized eigen vectors of A .

Q Reduce quadratic to canonical form
 $x_1^2 + 3x_2^2 + 3x_3^2 - 2x_2x_3$

$$\text{Let } Q = x_1^2 + 3x_2^2 + 3x_3^2 - 2x_2x_3$$

Then the matrix of quadratic form

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

To find Eigen values of A
char eqn $|A - \lambda I| = 0$

$$\lambda^3 - \text{tr}(A)\lambda^2 + |A_{11} + A_{22} + A_{33}| \lambda - \Delta = 0$$

$$\lambda^3 - 7\lambda^2 + [1 + 3 + 3]\lambda - 8 = 0$$

$$\lambda = 4, 2, 1$$

$\therefore$ The required canonical form is

$$Q = 4y_1^2 + 2y_2^2 + y_3^2$$

$$Q = 2x_1x_2 + 2x_1x_3 - 2x_2x_3$$

$$Q = x_1^2 + 2x_2^2 + x_3^2 - 2x_1x_2 + 2x_2x_3 + 6x_1x_3$$

$$Q = 5x_1^2 - 4x_1x_2 + 5x_2^2$$

Matrix $\begin{bmatrix} 0 & 1 & +1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}$

To find Eigen values of

$$|A - \lambda I| = 0$$

$$\lambda^3 - 0 + [0 + 1 + 1]\lambda + (-1 + 1) = 0$$

$$\lambda^3 - 3\lambda - 2 = 0$$

$$\lambda = 2, -1, -1$$

Canonical form is

$$Q = 2y_1^2 - y_2^2 - y_3^2$$

$$A = \begin{bmatrix} 1 & -1 & 3 \\ -1 & 2 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

$$\lambda^3 - 4\lambda^2 + (1 - 8 + 1)\lambda - (1 + 1(-4) + 3(-7))$$

The singular value decomposition (SVD)

The singular values of a matrix A are the square roots of the eigen values of A

SVD of A : For a matrix $A_{m \times n}$ of rank r

$\Rightarrow$

$$A = U \Sigma V^T$$

U is a $m \times m$ orthogonal matrix

V is a $n \times n$ " " "

Σ : $m \times n$ diagonal matrix

$$\Sigma = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}, D = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$$

singular values

Working rule

1. Compute $A^T A$ and find its eigen values.
 $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$
2. Find the corresponding eigen values.
 $v_1, v_2, \dots, v_n$
3. Normalize these vectors to get orthonormal vectors
 $\hat{v}_1, \hat{v}_2, \dots, \hat{v}_n$
4. Construct $V = [\hat{v}_1, \hat{v}_2, \hat{v}_3, \dots, \hat{v}_n]$, then find V^T

5. Compute singular values:

$$\sigma_i = \sqrt{\lambda_i}$$

6. Construct $D =$

7.

8.

9.

10.

Singular vectors

- * Columns of V : Right singular vectors
- * Columns of U : Left singular vectors

Note:

If all columns of V are not obtained from singular values, the remaining columns are chosen so that they are orthogonal to the existing column.

1. Construct a singular matrix $A = \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}$

$$\therefore \text{SVD of } A = U \Sigma V^T$$

$$A_{2 \times 2} = U_{2 \times 2} \Sigma_{2 \times 2} V^T$$

$$A = \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 2 & 2 \\ -1 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 8 & 2 \\ 2 & 5 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\text{Char eqn: } \lambda^2 - \text{tr}(A)\lambda + |A| = 0$$

$$\lambda^2 - 13\lambda + 36 = 0$$

$$\lambda = 9 \quad \lambda = 4$$

Eigen vectors:

When $\lambda = 9$

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x = 2y$$

$$X = \begin{bmatrix} 2 \\ 1 \end{bmatrix} = V_1$$

$$\lambda = 4$$

$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2x = y$$

$$x = 1, y = 2 \quad X = \begin{bmatrix} 1 \\ 2 \end{bmatrix} = V_2$$

$\therefore$ Eigen vectors $= V_1, V_2$

$$\hat{V}_1 = \frac{1}{\|V_1\|} V_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

$$\hat{V}_2 = \frac{1}{\|V_2\|} V_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$$

$$\therefore V = [\hat{V}_1, \hat{V}_2] = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix} = V^T$$

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{9} = 3$$

$$\sigma_2 = \sqrt{\lambda_2} = \sqrt{4} = 2$$

$$\therefore D = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \quad \Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

$$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} =$$

$$u_2$$

$$V = [u_1, u_2]$$

2. $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$ Ans.

$$A^T A = \begin{bmatrix} 1+4+4 & -9 \\ -9 & 9 \end{bmatrix} = \begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix}$$

$$\lambda^2 - \text{tr}(A)\lambda + |A| = 0$$

$$\lambda^2 - 18\lambda + 0 = 0$$

$$\lambda^2 = 18\lambda$$

$$\lambda = 18, 0$$

$$\lambda = 0$$

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x = y$$

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x = y$$

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda = 18$$

$$\begin{bmatrix} -9 & -9 \\ -9 & -9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} =$$

Final matrix of linear transformation

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$T(x, y) = \begin{pmatrix} x+y \\ -x \\ 3x-y \end{pmatrix}$$

$$\beta = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$\beta' = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$

Applying T on β'

$$T \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$c_1 + c_2 + c_3 = 2$$

$$c_1 + c_2 = 1$$

$$c_1 = 2$$

$$c_2 + 2 = 1$$

$$c_2 = -1$$

$$c_1 = 2 - 1 + c_3 = 2$$

$$c_3 = 1$$

$$T \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + (-1) \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$T\begin{pmatrix} 3 \\ 1 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 3 \\ 8 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$c_1 = 8$$

$$c_1 + c_2 = 3$$

$$c_1 + 8 = 3$$

$$c_2 = -5$$

$$c_1 + c_2 + c_3 = 4$$

$$8 - 5 + c_3 = 4$$

$$3 + c_3 = 4$$

$$c_3 = \underline{\underline{1}}$$

$$\therefore T\begin{pmatrix} 3 \\ 1 \end{pmatrix} = 8 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + (-5) \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$\therefore$ The required matrix of linear transformation

$$A = \begin{bmatrix} 2 & 6 \\ -1 & -5 \\ 1 & 1 \end{bmatrix}$$